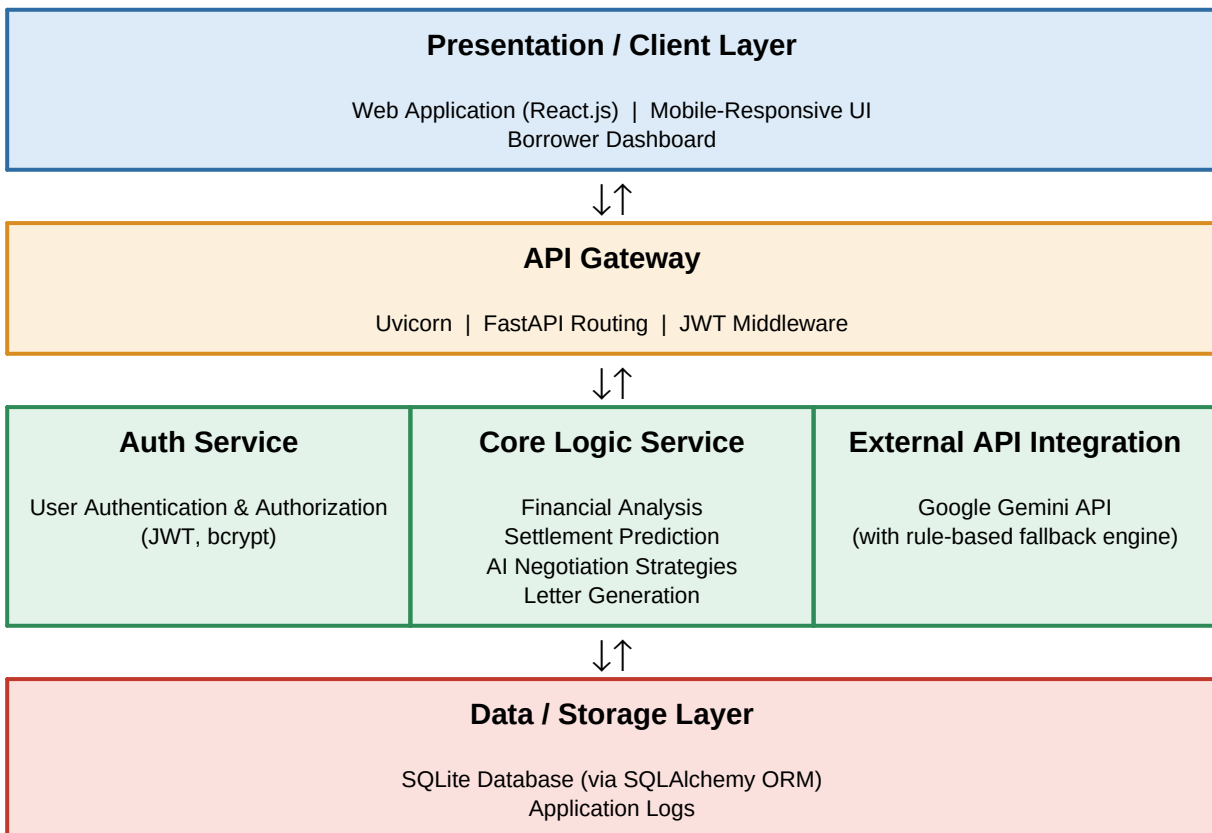


Solution Architecture

Date	5 July 2026
Team ID	
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Solution Architecture Diagram:

The diagram below illustrates the high-level architecture of the AI Powered Debt Relief & Financial Recovery Platform. It shows the flow and interaction between the Presentation Layer, Application Layer, and Data Layer along with key services and technologies used.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for borrowers to interact with the platform, including the web dashboard, loans, and settlement pages.	React.js, HTML5, CSS3, JavaScript (Vite)
API Gateway	Acts as a single entry point for all client requests, handles routing and JWT-based request validation.	Uvicorn, FastAPI
Core Logic Service	Contains core business services such as authentication, financial analysis, settlement prediction, and AI negotiation/letter generation.	Python (FastAPI), SQLAlchemy, Pydantic, python-jose, passlib
External API Integration	Integrates with the Google Gemini API for AI-driven negotiation strategy and letter generation, with a rule-based fallback.	Google Generative AI (Gemini API)
Database	Stores all application data including users, loans, financial profiles, settlement records, and AI interaction history.	SQLite, SQLAlchemy ORM